

Program 1:

```
import java.util.Scanner;
```

```
public class SimpleCalculator {
```

```
    public static void main(String[] args) {
```

```
        Scanner scanner = new Scanner(System.in);
```

```
        System.out.println("=== Simple Calculator ===");
```

```
        System.out.print("Enter first number: ");
```

```
        double num1 = scanner.nextDouble();
```

```
        System.out.print("Enter operator (+, -, *, /): ");
```

```
        char operator = scanner.next().charAt(0);
```

```
        System.out.print("Enter second number: ");
```

```
        double num2 = scanner.nextDouble();
```

```
        double result;
```

```
        switch (operator) {
```

```
            case '+':
```

```
                result = num1 + num2;
```

```
                System.out.println("Result: " + result);
```

```
                break;
```

```
            case '-':
```

```
                result = num1 - num2;
```

```
                System.out.println("Result: " + result);
```

```
                break;
```

```
            case '*':
```

```
                result = num1 * num2;
```

```
                System.out.println("Result: " + result);
```

```
                break;
```

```
            case '/':
```

```
                if (num2 != 0) {
```

```
                    result = num1 / num2;
```

```
                    System.out.println("Result: " + result);
```

```
                } else {
```

```
                    System.out.println("Error: Division by zero is not allowed.");
```

```
                }
```

```
                break;
```

```
            default:
```

```
                System.out.println("Error: Invalid operator.");
```

```
        }
```

```

        scanner.close();
    }
}

```

Program 2:

```

public class ReverseString {
    public static void main(String[] args) {
        String str = "Hello World";
        String reversed = "";

        for (int i = str.length() - 1; i >= 0; i--) {
            reversed += str.charAt(i);
        }

        System.out.println("Original String: " + str);
        System.out.println("Reversed String: " + reversed);
    }
}

```

Program 3:

```

import java.util.Scanner;

public class ReadArray {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        // Read number of elements
        System.out.print("Enter number of elements: ");
        int n = scanner.nextInt();

        int[] arr = new int[n];

        // Read elements
        System.out.println("Enter " + n + " elements:");
        for (int i = 0; i < n; i++) {
            arr[i] = scanner.nextInt();
        }

        // Display the array
        System.out.println("Array elements are:");
        for (int i = 0; i < n; i++) {
            System.out.print(arr[i] + " ");
        }

        scanner.close();
    }
}

```

}